FOR EXTERNAL EXAMINER (date of this version: 12/4/2023)

UNIVERSITY OF EDINBURGH

COLLEGE OF SCIENCE AND ENGINEERING

SCHOOL OF INFORMATICS

INFR11171 DATA ANALYTICS WITH HIGH PERFORMANCE COMPUTING

Tuesday 2nd May 2023

13:00 to 15:00

INSTRUCTIONS TO CANDIDATES

1. **ALL QUESTIONS ARE COMPULSORY WITH A TOTAL OF 75-MARKS.**
2. **DIFFERENT SUB-QUESTIONS MAY HAVE DIFFERENT NUMBERS. Take note of this in allocating time to questions.**
3. **CALCULATORS MAY BE USED IN THIS EXAMINATION.**

EPCC Courses

Convener: Dr Andreas Pieris
External Examiner: Prof Matt Probert

THIS EXAMINATION WILL BE MARKED ANONYMOUSLY

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1. As part of a national reporting scheme, every hospital in Scotland has been told to provide information on patient admissions since 2010 and on an ongoing basis into the future. You work at the reporting centre and have been tasked with cleaning the data so it can be loaded into a database for different analysis tasks. Each hospital provides one CSV file per month; however, you notice that some files are missing. You are also told that some data have been entered by hand.

   Each file is supposed to contain the following variables:

   - *unique_patient_id* e.g., "03-05-92-0124",
   - *hospital_location_code* e.g., "EDI004",
   - *admission_date*, *discharge_date*, and *length_of_stay*,
   - *admission_type*, which should be one of "Emergency", "Elective", or "Transfer",
   - *department_code* e.g.,"RAD", "ORTHO", "ENT".

   (a) The dataset is an example of Operational data which has not been gathered for a specific analysis. Explain why it is vital to clean this data before it is made available for analysis, and state two resources you could use to assist with the cleaning. [*3 marks*]

   (b) Name the technique which could be applied to convert a categorical variable for use by numerical models, and show how it would be applied to the following data sample. [*2 marks*]

Row	*admission_type*
1	Emergency
2	Emergency
3	Transfer

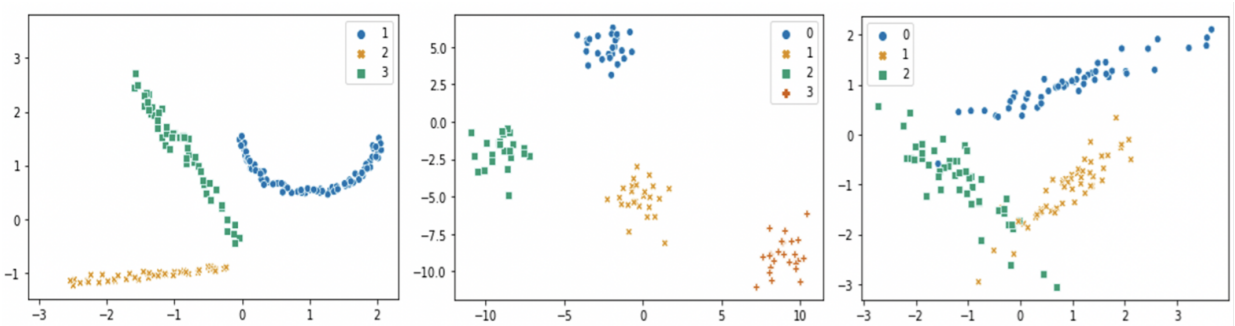
   (c) Create a plan for cleaning the dataset above, providing specific examples of tasks at each step in the Data Cleaning workflow. You should assume that HPC is **not** required for this task. [*12 marks*]

   (d) Briefly describe how you could verify the following statement using the cleaned data: "For each hospital, the average length of stay per year should be much higher for transfers than any other admission type". [*2 marks*]

   (e) A researcher wishes to train a Random Forest classifier using the cleaned data to predict future admission rates. Explain what the terms "bootstrap aggregation" and "out of bag (OOB) error" mean in this context. [*2 marks*]

   (f) Describe an experiment you could perform to determine the optimal number of trees to include in a Random Forest classifier. [*4 marks*]

2. (a) The plots given below show the results of different clustering methods used on 3 different datasets. Looking at these, which of k-means, Gaussian Mixtures and DBSCAN is most likely to have produced each clustering output? Justify your answer based on the properties and underlying assumptions of each method. *[3 marks]*



(b) Let x be a datapoint with x_1, ..., x_9 denoting its closest neighbours, with their Euclidean distances from x listed below. Explain how DBSCAN would classify x, i.e., core point, density reachable or outlier, if parametrised with $\epsilon = 0.12$ and $minPts = 3$, and briefly justify why.

$$\begin{aligned} D(x, x_1) &= 0.4, D(x, x_2) = 0.08, D(x, x_3) = 0.15, D(x, x_4) = 0.2, \\ D(x, x_5) &= 0.3, D(x, x_6) = 0.22, D(x, x_7) = 0.06, D(x, x_8) = 0.11, D(x, x_9) = 0.1 \end{aligned}$$

[2 marks]

(c) Your local events organiser wants to determine the success of its events, and whether a future event will sell out. For this, you're given a CSV file with three variables: Y : showing if the event was sold out or not; $X1$: the type of the event and $X2$: whether it took place during the weekend or not. After processing the data you bring them together in the below format:

	music event	dancing event
sold out	100	90
not sold out	50	20

	weekday	weekend
sold out	80	110
not sold out	40	30

i. They plan to organise a dancing event on the last Thursday of May. Use a Naive Bayes classifier to answer whether it is going to be sold out or not. Show all your working using appropriate notation.

ii. They consider organising a poetry writing event during June's first weekend and they want to know if it's going to be sold out. Can you answer this using the provided data and/or the trained classifier? If yes, show and explain how but without making the actual calculation. If not, explain why and suggest how you could approach it.

[*11 marks*]

(d) Your electricity provider recently launched a new service on their website and the company now wishes to study their customers' response and whether this launch was successful. Let Y be the random variable indicating whether a customer responded positively or not, where $Y = 1$ indicates a positive response and $Y = 0$ otherwise. A sample of $n = 2000$ customers was randomly drawn and response status was recorded resulting to the sample: $y = 0, 0, 0, 0, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0, \ldots, 0, 0, 1, 0, 1, 1, 0, 0, 0, 0, 0, 0, 1, 1.$

i. What is an appropriate distribution for modelling Y and why? Provide the parameter(s) of the distribution and what they illustarte.

ii. You are given additional information for each customer, i.e., X_1 : contract type, X_2 : length of tenure and X_3 : average monthly consumption. With this additional information you are asked to model the probability of a customer responding positively to the offer and you decide to use logistic regression.

- Fully define the hypothesis model and explain how you'd use your trained model in order to predict if a customer is going to respond positively before they are presented with the offer. Provide any equation that you may have to use.

iii. Would linear regression be an appropriate model instead of logistic regression for this situation? Why?

[*9 marks*]

3. You are tasked with analysing sales data for a musical instrument retailer, **Strings-n-Keys**. The retailer has 5 years of sales data, from 10 physical stores and an online store, comprising 600 files for monthly in-person sales, and 260 files for weekly online sales. Example data are shown below.

***In-person Sales**, showing line 1 from 4 CSV files.*

date	**store**	**department**	**type**	**maker**	**model**	**price**
01-01-18	Edinburgh	Guitars	Guitar	Fender	**Strat**	999.00
01-01-18	Glasgow	Drums	Drum kit	Gretsch	BC50	1849.00
01-02-18	Dundee	Guitars	Guitar	Gibson	LP	1399.00
01-02-18	Bristol	Keys	Keyboard	Nord	X300	2199.00

***Online Sales**, showing 3 lines from a single CSV file.*

datetime	**department**	**type**	**maker**	**model**	**price**	**shipping**
20-01-2019 12:36:21	Keys	Keyboard	Casio	88Pro	1299.00	30.00
20-01-2019 12:42:19	Drums	Drum stool	Gnu	Red15	149.00	30.00
20-01-2019 13:01:09	Recording	Interface	UA	Volt1	114.99	10.00

The client would like to see monthly sales and yearly sales for each of 10 departments across all stores and to predict future sales. To complete this work, you have at your disposal a small HPC machine featuring 8 compute nodes, each with 12 CPU cores and 96 GB RAM, that uses the SLURM job scheduler.

(a) You begin by designing a single table combining in-person and online sales, assuming all stores have the same departments.

   i. What columns are required, how should the table be indexed and what data types should be used?

   ii. What method should be used to combine time-based data and how should it be aggregated?

[*5 marks*]

(b) To handle the large amount of data, you need to make full use of a HPC machine and use either **Spark** (using PySpark) or **Dask**.

   i. How do **Spark** and **Dask** setups differ in terms of the way the SLURM job scheduler is used and how are computation tasks executed?

   ii. To load the data, **Spark** and **Dask** can read in lists of paths to CSV files. Describe what `spark.read.csv()` and `dask.dataframe.read_csv()` do and what they produce.

[*6 marks*]

(c) The **Spark Pipeline** diagram below represents a process to model the **String-n-Keys** data.

i. Which of these stages are *Transformers* and which are *Estimators*? If a stage is a *Transformer*, state, with reasoning, whether it performs a *narrow* or a *wide* transformation.

ii. Once the **Spark Pipeline** has been processed, *Estimator* stages become *Transformer* stages in a now trained **PipelineModel** object. What can be done with the **PipelineModel** and what additional stage type can be added? [*4 marks*]

(d) In the example below, the `dask.delayed()` function is used as a decorator to simple arithmetic functions to build a **Dask** task graph, where "`result`" is the final task graph.

```
1  import dask
2
3  @dask.delayed
4  def add(x, y):
5          return x + y
6
7  @dask.delayed
8  def divide(x, y):
9          return x / y
10
11 @dask.delayed
12 def square(x):
13         return x * x
14
15 a = square(4)
16 b = square(2)
17 c = add(a, b)
18 result = divide(c, 2)
19
20 result.visualize()
21 result.compute()
```

i. Draw the task graph that is output by line 21, "`result.visualize()`", highlighting which tasks run in parallel and which run in serial. Use circles to represent operations and rectangles to represent values.

ii. What is the result of line 22, "`result.compute()`"?

iii. In terms of the **String-n-Keys** data, a larger and more complex task graph would be generated. What is the initial number of partitions used and why? What would be the most efficient way to repartition the data, and why would this improve efficiency? [*6 marks*]

(e) Assume you have now modelled the **String-n-Keys** data with a multiple linear regression model, where stores and departments are represented as independent variables.

i. You want to compare yearly sales of all physical sales against online sales. For such a comparison, what operations would be required to re-

organise the data and to produce a scatter plot of observed data showing the fitted linear model as an overlay?

ii. How would you predict sales figures for each department of the Edinburgh store for December 2023?

[*4 marks*]